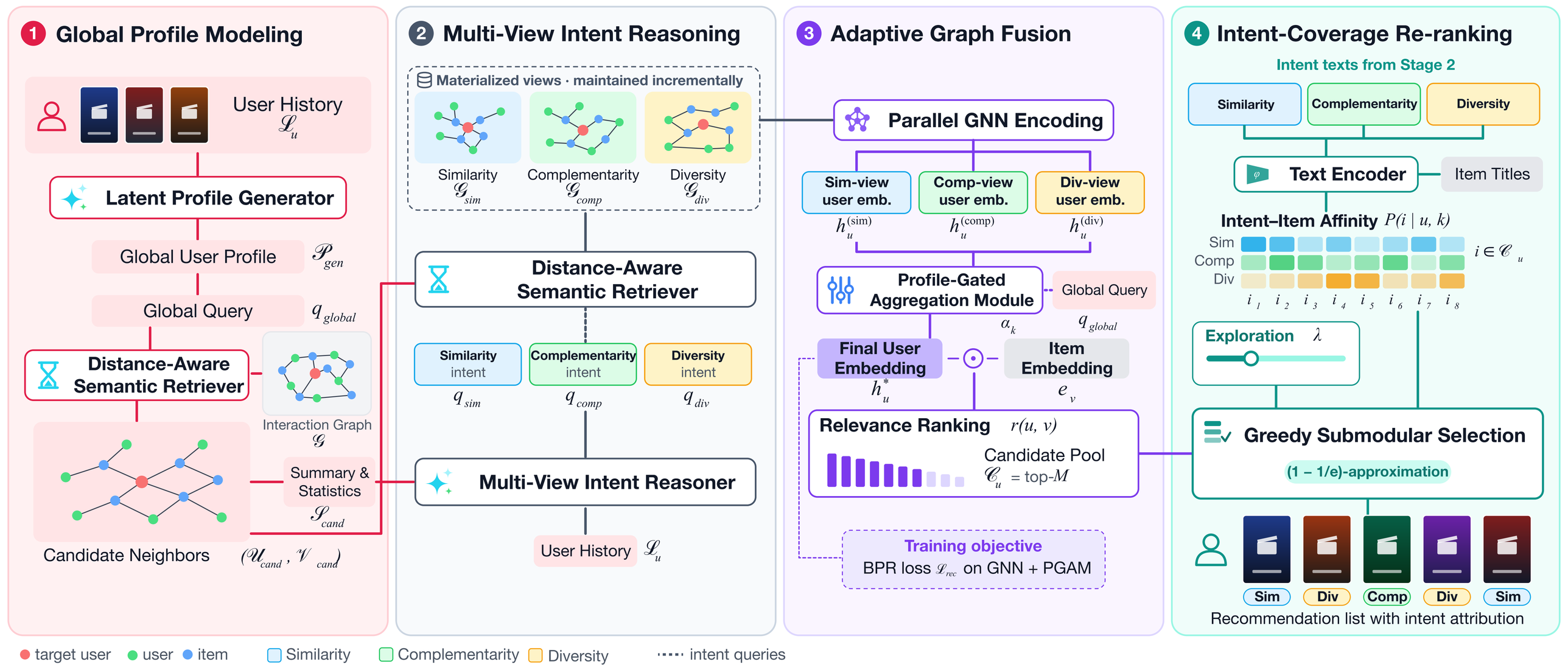

1 Global Profile Modeling
User History $\mathcal{L}_u$
Latent Profile Generator
Global User Profile $\mathcal{P}_{gen}$
Global Query q_{global}
Distance-Aware Semantic Retriever
Interaction Graph $\mathcal{G}$
Summary & Statistics $\mathcal{S}_{cand}$
Candidate Neighbors $(\mathcal{U}_{cand}, \mathcal{V}_{cand})$
2 Multi-View Intent Reasoning
Materialized views · maintained incrementally
Similarity $\mathcal{G}_{sim}$
Complementarity $\mathcal{G}_{comp}$
Diversity $\mathcal{G}_{div}$
Distance-Aware Semantic Retriever
Similarity intent q_{sim}
Complementarity intent q_{comp}
Diversity intent q_{div}
Multi-View Intent Reasoner
User History $\mathcal{L}_u$
3 Adaptive Graph Fusion
Parallel GNN Encoding
Sim-view user emb. $h_u^{(sim)}$
Comp-view user emb. $h_u^{(comp)}$
Div-view user emb. $h_u^{(div)}$
Profile-Gated Aggregation Module
Global Query q_{global}
α_k
Final User Embedding h_u^*
Item Embedding e_v
Relevance Ranking $r(u, v)$
Candidate Pool $\mathcal{C}_u$ = top-M
Training objective
BPR loss $\mathcal{L}_{rec}$ on GNN + PGAM
4 Intent-Coverage Re-ranking
Intent texts from Stage 2
Similarity
Complementarity
Diversity
Text Encoder
Item Titles
Intent–Item Affinity $P(i \mid u, k)$
Sim
Comp
Div
$i \in \mathcal{C}_u$
i_1 i_2 i_3 i_4 i_5 i_6 i_7 i_8
Exploration λ
Greedy Submodular Selection
(1 – 1/e)-approximation
Sim
Div
Comp
Div
Sim
Recommendation list with intent attribution
target user
user
item
Similarity
Complementarity
Diversity
intent queries